Computational Work 3

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Course: Computational Matrix Methods

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Abstract

This work investigates two least-squares problems using data derived from the student's enrollment number. First, an overdetermined linear system $A\mathbf{x} \approx \mathbf{b}$, with $A \in \mathbb{R}^{5\times 4}$, is solved using the QR factorization $A = QR$. The least-squares solution is obtained by solving the resulting upper triangular system. The existence of a unique solution is discussed in terms of the rank of the coefficient matrix.

Second, a quadratic polynomial is fitted to six experimental data points using the least-squares formulation. The associated design matrix is constructed, and the polynomial coefficients are determined computationally. The quality of the fit is assessed by plotting the experimental data and the fitted curve and calculating the mean squared error.

The numerical implementation explicitly constructs the principal linear algebra algorithms, avoiding high-level routines for QR factorization and least-squares solutions. This approach connects the mathematical foundations of least squares with their computational implementation and numerical evaluation.

Keywords: QR factorization; least squares; quadratic polynomial fitting; numerical linear algebra.

1 Problem Formulation

This work addresses two computational problems involving least-squares methods. The numerical data are constructed from the eight digits of the student's enrollment number. Any zero digit is replaced by one before constructing the matrices and vectors.

1.1 Problem 1: QR Factorization and Linear Least Squares

Consider the overdetermined linear system

$$A\mathbf{x} \approx \mathbf{b}, \tag{1}$$

where $A \in \mathbb{R}^{5\times 4}$, $\mathbf{x} \in \mathbb{R}^4$, and $\mathbf{b} \in \mathbb{R}^5$. The coefficient matrix and right-hand-side vector are defined by

$$A = \begin{bmatrix} m_1+2 & m_2 & -m_3 & m_4 \\ m_2+1 & -m_3 & m_4+2 & m_5 \\ m_3+2 & -m_4 & m_5+1 & -m_6 \\ -m_4 & m_5+2 & m_6 & m_7 \\ -m_5 & m_6+1 & m_7+2 & m_8 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} m_1+m_2 \\ -m_3 \\ m_4+m_5 \\ -m_6 \\ m_7+m_8 \end{bmatrix}. \tag{2}$$

The first objective is to compute the reduced QR factorization

$$A = QR, \tag{3}$$

where $Q \in \mathbb{R}^{5\times 4}$ has orthonormal columns and $R \in \mathbb{R}^{4\times 4}$ is upper triangular.

The least-squares solution is defined as

$$\mathbf{x}^* = \arg\min_{\mathbf{x}\in\mathbb{R}^4} \|A\mathbf{x} - \mathbf{b}\|_2^2. \tag{4}$$

The uniqueness of this solution depends on the column rank of A. In particular, the solution is unique if and only if $\text{rank}(A) = 4$. This condition will be assessed computationally for the given enrollment number.

1.2 Problem 2: Quadratic Least-Squares Fitting

Consider six experimental data points (x_i, y_i), where

$$x_i = i, \qquad y_i = m_i \bmod 5, \qquad i = 1, \ldots, 6. \tag{5}$$

The objective is to approximate these data using a quadratic polynomial

$$p(x) = ax^2 + bx + c. \tag{6}$$

Defining the coefficient vector $\boldsymbol{\theta} = [a, b, c]^T$, the fitting problem can be expressed as

$$X\boldsymbol{\theta} \approx \mathbf{y}, \tag{7}$$

where

$$X = \begin{bmatrix} 1^2 & 1 & 1 \\ 2^2 & 2 & 1 \\ 3^2 & 3 & 1 \\ 4^2 & 4 & 1 \\ 5^2 & 5 & 1 \\ 6^2 & 6 & 1 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \end{bmatrix}. \tag{8}$$

The polynomial coefficients are obtained by solving

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}\in\mathbb{R}^3} \|X\boldsymbol{\theta} - \mathbf{y}\|_2^2. \tag{9}$$

Since the six values of x_i are distinct, the three columns of X are linearly independent. Therefore, $\text{rank}(X) = 3$, and the least-squares solution is unique.

Finally, the quality of the fitted polynomial is evaluated by plotting the experimental points and

the fitted curve and by computing the mean squared error (MSE):

$$\text{MSE} = \frac{1}{6}\sum_{i=1}^{6}\left[y_i - p(x_i)\right]^2. \tag{10}$$

2 Methodology

2.1 Data Construction

The numerical data are generated according to the instructions provided in the assignment. The student's enrollment number is converted into eight integer digits, replacing every zero with one. These digits are then used to construct the coefficient matrix A, the vector $\mathbf{b}$, and the experimental observations (x_i, y_i).

The original data-generation expressions are preserved. NumPy is used for the array construction specified in the assignment, whereas the principal linear algebra operations are implemented explicitly using Python loops. In particular, no predefined routines for QR factorization, least-squares solutions, or polynomial fitting are used.

2.2 QR Factorization by Modified Gram–Schmidt

Let $B \in \mathbb{R}^{m\times n}$, with $m \geq n$, be a matrix with linearly independent columns. The QR factorization expresses B as

$$B = QR, \tag{11}$$

where $Q \in \mathbb{R}^{m\times n}$ has orthonormal columns, $Q^TQ = I_n$, and $R \in \mathbb{R}^{n\times n}$ is upper triangular.

The factorization is computed using the modified Gram–Schmidt (MGS) algorithm. This method constructs an orthonormal basis for the column space of B by successively removing the components of each column in the directions of previously computed basis vectors.

Denote the j-th column of B by $\mathbf{b}_j$. Initialize

$$\mathbf{v}_j^{(0)} = \mathbf{b}_j. \tag{12}$$

For $i = 1, \ldots, j-1$, compute

$$r_{ij} = \mathbf{q}_i^T\mathbf{v}_j^{(i-1)}, \tag{13}$$

$$\mathbf{v}_j^{(i)} = \mathbf{v}_j^{(i-1)} - r_{ij}\mathbf{q}_i. \tag{14}$$

The next diagonal entry of R and the corresponding column of Q are then obtained as

$$r_{jj} = \|\mathbf{v}_j^{(j-1)}\|_2, \tag{15}$$

$$\mathbf{q}_j = \frac{\mathbf{v}_j^{(j-1)}}{r_{jj}}. \tag{16}$$

The algorithm is applied first to the matrix A and subsequently to the design matrix X. The

resulting factors are used to solve both least-squares problems. During implementation, a diagonal entry of R that is numerically negligible is treated as an indication of possible rank deficiency; the corresponding case must be reported rather than dividing by that entry.

2.3 Selection of the Orthogonalization Method

Several methods can compute the QR factorization. The choice depends on numerical stability, implementation complexity, and the objectives of the computational assignment.

Table 1: Comparison of methods for computing the QR factorization.

Method	**Advantages**	**Limitations**
Classical Gram–Schmidt	Simple formulation and straightforward implementation.	Can lose orthogonality when the columns are nearly linearly dependent.
Modified Gram–Schmidt	Explicit construction of the orthonormal basis; generally preserves orthogonality better than the classical variant and is convenient for implementing the algorithm from first principles.	Can still lose orthogonality for ill-conditioned matrices; it is not generally as numerically robust as Householder QR.
Householder reflections	Highly reliable numerical behavior and a standard choice for dense QR factorizations.	Requires implementing and managing reflection transformations, making the algorithm less direct to explain from first principles.
Givens rotations	Useful for exploiting sparsity and updating factorizations when individual rows or entries change.	Requires managing a sequence of rotations, which is unnecessary for the small dense matrices considered here.

The modified Gram–Schmidt method is selected primarily for its transparent computational structure and suitability for a from-scratch implementation. It provides a direct illustration of orthogonalization while improving numerical behavior relative to classical Gram–Schmidt. Householder QR would generally be preferable when numerical robustness is the primary concern, particularly for larger or ill-conditioned problems [1, 2, 3].

The choice is therefore pedagogical and computational rather than a claim that modified Gram–Schmidt is universally the most stable QR algorithm.

2.4 Least-Squares Solution Using QR

Consider the overdetermined system

$$B\mathbf{z} \approx \mathbf{c}, \tag{17}$$

where B has full column rank. The least-squares solution minimizes the squared Euclidean residual:

$$\mathbf{z}^* = \arg\min_{\mathbf{z}\in\mathbb{R}^n} \|B\mathbf{z} - \mathbf{c}\|_2^2. \tag{18}$$

Substituting $B = QR$ and using the orthonormality of the columns of Q gives

$$\begin{aligned} \|B\mathbf{z} - \mathbf{c}\|_2^2 = \|R\mathbf{z} - Q^T\mathbf{c}\|_2^2 \\ + \|(I - QQ^T)\mathbf{c}\|_2^2. \end{aligned} \tag{19}$$

The second term is independent of $\mathbf{z}$. Consequently, the least-squares solution is obtained by solving the upper triangular system

$$R\mathbf{z} = Q^T\mathbf{c}. \tag{20}$$

This system is solved by back substitution, starting from the last equation:

$$z_i = \frac{d_i - \sum_{j=i+1}^{n} R_{ij} z_j}{R_{ii}}, \qquad i = n, n-1, \ldots, 1, \tag{21}$$

where $\mathbf{d} = Q^T\mathbf{c}$.

For the first problem, $B = A$, $\mathbf{c} = \mathbf{b}$, and $\mathbf{z} = \mathbf{x}$. For the second problem, $B = X$, $\mathbf{c} = \mathbf{y}$, and $\mathbf{z} = [a, b, c]^T$.

The QR-based approach is preferred to solving the normal equations $B^TB\mathbf{z} = B^T\mathbf{c}$. For a full-column-rank matrix, the normal equations satisfy $\kappa_2(B^TB) = \kappa_2(B)^2$, potentially amplifying the effects of poor conditioning. The QR formulation avoids explicitly forming B^TB and reduces the least-squares problem to a triangular solve [1, 3].

2.5 Numerical Verification and Error Evaluation

The computed QR factors are assessed using two quantities. The relative factorization error is defined as

$$\eta_{\text{QR}} = \frac{\|B - QR\|_F}{\|B\|_F}, \tag{22}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. The orthogonality error is defined as

$$\eta_Q = \|Q^TQ - I_n\|_F. \tag{23}$$

Both quantities are evaluated using explicitly implemented matrix operations.

For the first problem, the residual norm

$$\rho_A = \|A\mathbf{x}^* - \mathbf{b}\|_2 \tag{24}$$

is calculated to assess the least-squares solution. The uniqueness of the solution is examined through the column rank of A and the diagonal entries of R.

For the quadratic fit, the predicted values are calculated from

$$\widehat{y}_i = ax_i^2 + bx_i + c, \tag{25}$$

and the mean squared error is computed as

$$\text{MSE} = \frac{1}{6}\sum_{i=1}^{6}(y_i - \widehat{y}_i)^2. \tag{26}$$

Finally, the experimental observations and the fitted polynomial are plotted together. The numerical algorithms are implemented without predefined linear algebra or polynomial-fitting routines; a plotting library is used only to visualize the results.

The complete computational implementation, including the data-generation code and the routines for QR factorization, back substitution, error evaluation, and plotting, is provided in the appendix.

3 Results Discussion

3.1 QR Factorization and Linear Least-Squares Solution

The modified Gram–Schmidt algorithm was applied to the overdetermined system defined in Section 1. The computed factors satisfy $A \approx QR$, where $Q \in \mathbb{R}^{5\times 4}$ has approximately orthonormal columns and $R \in \mathbb{R}^{4\times 4}$ is upper triangular.

The computed matrices, rounded to six decimal places, are

$$Q \approx \begin{bmatrix} 0.295599 & 0.242348 & -0.209363 & 0.159387 \\ 0.197066 & -0.200200 & 0.322900 & 0.898629 \\ 0.492665 & 0.140811 & 0.781674 & -0.352688 \\ -0.098533 & 0.938737 & -0.000348 & 0.202929 \\ -0.788263 & 0.011495 & 0.490803 & 0.038631 \end{bmatrix}, \tag{27}$$

and

$$R \approx \begin{bmatrix} 10.148892 & -4.926646 & -3.251587 & -4.236916 \\ 0 & 10.135490 & 2.859317 & 4.846950 \\ 0 & 0 & 13.048045 & 2.480397 \\ 0 & 0 & 0 & 10.020142 \end{bmatrix}. \tag{28}$$

The quality of the factorization was assessed using the relative reconstruction error and the orthogonality error. The computed values were

$$\eta_{\text{QR}} = 1.0683 \times 10^{-16}, \qquad \eta_Q = 4.5180 \times 10^{-16}. \tag{29}$$

Both indicators are close to the expected scale of floating-point rounding errors for this computation. These results support the numerical accuracy of the factorization and the orthogonality of the computed columns of Q.

The least-squares solution was obtained by solving $R\mathbf{x}^* = Q^T\mathbf{b}$ using back substitution. The resulting solution, rounded to nine decimal places, is

$$\mathbf{x}^* \approx \begin{bmatrix} -0.404938762 \\ -0.040220457 \\ 0.992366790 \\ -0.568506366 \end{bmatrix}. \tag{30}$$

The Euclidean norm of the residual was

$$\|\mathbf{A}\mathbf{x}^* - \mathbf{b}\|_2 \approx 7.677354. \tag{31}$$

The nonzero residual indicates that the original system is inconsistent. This is expected for an overdetermined least-squares problem and does not, by itself, indicate a failure of the numerical method. Instead, the solution minimizes the residual norm among all vectors in $\mathbb{R}^4$.

To assess the optimality of the computed solution, the norm of the residual projected onto the column directions of A was also evaluated. The result was

$$\left\|A^T(A\mathbf{x}^* - \mathbf{b})\right\|_2 \approx 6.1020 \times 10^{-14}. \tag{32}$$

This value is numerically small and is consistent with the least-squares optimality condition $A^T(A\mathbf{x}^* - \mathbf{b}) = \mathbf{0}$. Hence, the computed residual is approximately orthogonal to the column space of A.

Finally, the diagonal entries of R were compared with the scale-dependent numerical tolerance used by the implementation. All four diagonal entries were substantially larger than this tolerance, supporting the numerical assessment that A has full column rank. Therefore, the least-squares solution is unique.

These results validate the use of the modified Gram–Schmidt factorization and back substitution for the matrix considered in this experiment. Nevertheless, modified Gram–Schmidt without column pivoting is not a universally reliable rank-revealing method for severely ill-conditioned matrices. Householder QR or rank-revealing factorizations may be preferable when stronger numerical robustness is required [1, 3].

3.2 Quadratic Least-Squares Fitting

The second experiment considered the least-squares approximation of six observations by a polynomial of degree at most two. The design matrix $X \in \mathbb{R}^{6\times 3}$ was constructed using the columns associated with x^2, x, and the constant term. The coefficient vector $\boldsymbol{\theta} = [a, b, c]^T$ was computed using the reduced QR factorization and back substitution.

The computed coefficients were

$$\boldsymbol{\theta}^* \approx \begin{bmatrix} 1.9739 \times 10^{-15} \\ 0.4000 \\ 0.6000 \end{bmatrix}. \tag{33}$$

Thus, the fitted polynomial is

$$p(x) \approx 0.4000x + 0.6000, \tag{34}$$

because the estimated quadratic coefficient is numerically negligible. Although a quadratic model was specified, the resulting least-squares polynomial is effectively linear at the displayed precision. This behavior is a property of the data and does not indicate a failure of the numerical algorithm.

Table 2 compares the observations with the fitted values and the corresponding residuals.

Table 2: Observations, fitted values, and residuals.

x_i	y_i	$\widehat{y}_i$	$y_i - \widehat{y}_i$
1	1	1.0	0.0
2	1	1.4	-0.4
3	3	1.8	1.2
4	1	2.2	-1.2
5	3	2.6	0.4
6	3	3.0	0.0

The residual vector has Euclidean norm

$$\|\mathbf{y} - X\boldsymbol{\theta}^*\|_2 \approx 1.7889. \tag{35}$$

The mean squared error is

$$\text{MSE} = \frac{1}{6}\sum_{i=1}^{6}(y_i - \widehat{y}_i)^2 = \frac{3.2}{6} \approx 0.5333. \tag{36}$$

The residual is nonzero because the fitted polynomial does not pass through every observation. This is expected in a least-squares approximation and does not imply that the computed minimizer is inaccurate. The MSE quantifies the discrepancy between the data and the fitted model, whereas the numerical indicators assess the accuracy of the computational procedure.

The QR factorization produced a relative reconstruction error of 7.3017×10^{-17}, while the orthogonality error was 3.9468×10^{-15}. The least-squares optimality indicator, $\|X^T(X\boldsymbol{\theta}^* - \mathbf{y})\|_2$, was 2.3055×10^{-14}. These values are consistent with a numerically accurate solution for this problem.

The uniqueness of the solution follows from the full column rank of X. Indeed, if $X\boldsymbol{\theta} = \mathbf{0}$, then the polynomial $ax^2 + bx + c$ vanishes at the six distinct values $x_i = 1, \ldots, 6$. A nonzero polynomial of degree at most two cannot have six distinct roots. Therefore, $\boldsymbol{\theta} = \mathbf{0}$, so $\text{rank}(X) = 3$ and the least-squares minimizer is unique.

Figure 1 presents the observations and the fitted polynomial. The nearly straight fitted curve is consistent with the numerically negligible quadratic coefficient.

Since the observations are constructed from the enrollment digits as prescribed by the assignment, the analysis concerns the numerical behavior of the fitting procedure rather than an empirical conclusion about predator–prey dynamics.

References

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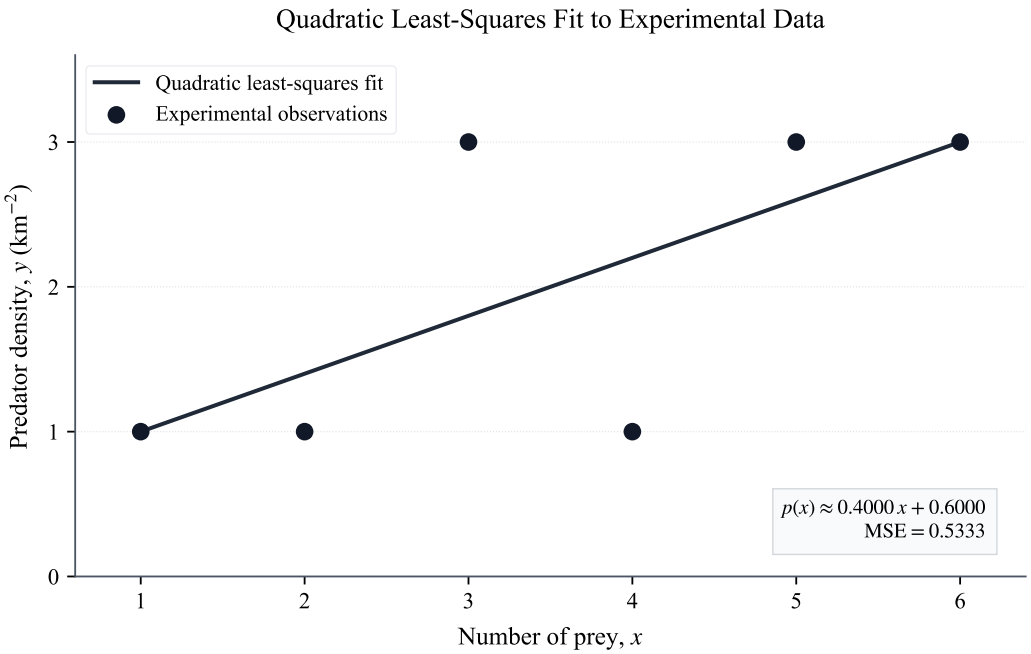


Figure 1: Experimental observations and the quadratic least-squares fitted curve.

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